ENGRAMABENCH: Evaluating Long-Term Conversational Memory with Structured Graph Retrieval

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Abstract

Large language model assistants are increasingly expected to retain, connect, and reason over information accumulated across many sessions. We introduce ENGRAMABENCH, a benchmark for long-term conversational memory built around five canonical personas, one hundred multi-session conversations, and one hundred fifty queries spanning factual recall, cross-space integration, temporal reasoning, adversarial abstention, and emergent synthesis. We evaluate ENGRAMA, a proprietary graph-structured memory system, against two strong baselines: GPT-4o full-context prompting and MEM0, an open-source memory system using vector retrieval over extracted memories. All three systems use the same answering model (GPT-4o), isolating the effect of memory architecture. GPT-4o full-context achieves the highest composite score (0.6186), while ENGRAMA scores 0.5367 globally but is the only system to outperform full-context prompting on `cross_space` reasoning (0.6532 vs. 0.6291), the benchmark slice most diagnostic of structured long-term memory. MEM0 is cheapest at query time but substantially weaker overall (0.4809). Ablations reveal that the components responsible for ENGRAMA's cross-space advantage trade off against global composite score, exposing a real systems-level tension between structured memory specialization and aggregate benchmark optimization. These results suggest that structured memory does not yet replace full-context prompting universally, but it can already recover a meaningful advantage where memory organization itself matters.

1 Introduction

Large language model assistants are increasingly expected to behave as if they remember. Users do not interact with them as one-shot question answering systems, but as persistent collaborators that accumulate context over days, weeks, and months. In this regime, a useful assistant must do more than handle long prompts. It must retain information from prior interactions, recover it when relevant, update it when circumstances change, and connect facts that were originally discussed in different parts of a user's life. These are fundamentally long-term memory problems.

One obvious way to address this problem is to provide the model with as much prior conversation history as possible at answer time. Recent long-context models make this strategy increasingly viable, and in many settings it remains surprisingly strong. However, full-context prompting has two practical limitations. First, it scales poorly in serving cost as conversation histories grow. Second, it treats memory as transcript inclusion rather than as a structured representation of what the assistant has learned. As a result, it remains unclear whether full-context prompting is the right long-term solution.

This tension motivates two questions. First, *what kinds of memory behavior actually matter* in long-term conversational systems? Second, *when does structured memory provide an advantage over brute-force context inclusion or flat retrieval over extracted facts?* Existing benchmarks and systems only partially answer these questions. Some focus on short-horizon personalization or human-human dialogue; others do not isolate cross-space memory integration, temporal change, or abstention under long multi-session interaction. And many memory-augmented systems are difficult to analyze because they conflate memory construction, retrieval, and answer generation into a single opaque pipeline.

In this paper, we present two contributions aimed at making this problem more legible. First, we introduce ENGRAMABENCH, a benchmark for long-term conversational memory built around five canonical personas, one hundred multi-session conversations, and one hundred fifty queries spanning factual recall, cross-space integration, temporal reasoning, adversarial abstention, and emergent synthesis. Second, we evaluate ENGRAMA, a proprietary graph-structured memory system, on this benchmark and compare it to two strong alternatives: **GPT-4o full-context**, which injects the entire conversation history at answer time, and MEM0, an open-source memory system based on vector retrieval over extracted memories.

The resulting picture is more nuanced than a simple ranking. Full-context prompting achieves the highest composite score (**0.6186**), but ENGRAMA is the only system to outperform it on `cross_space` reasoning (**0.6532** vs. **0.6291**), while MEM0 trails both substantially (**0.4809**). Ablations reveal that the components driving ENGRAMA's cross-space strength are not the same ones that maximize aggregate score, exposing a design tension that helps explain why memory architectures can appear uniformly promising in smaller evaluations but reveal structured costs under broader benchmarking.

These findings reframe the central question: not whether to remember more history, but how to remember it in a form that remains useful across heterogeneous user narratives and reasoning demands.

Our contributions are as follows:

- We introduce ENGRAMABENCH, a benchmark for long-term conversational memory with explicit coverage of `single_space`, `cross_space`, `temporal_cross_space`, `adversarial`, and `emergent_insight` behavior.
- We present ENGRAMA, a graph-structured memory system for long-term conversational retrieval and reasoning, described at a high level sufficient for empirical analysis.
- We provide a controlled comparison among ENGRAMA, GPT-4o full-context, and MEM0, showing that full-context prompting wins globally, while ENGRAMA uniquely wins on `cross_space` reasoning at substantially lower query-time cost.
- We analyze internal design tradeoffs through ablations, separating the mechanisms that improve cross-space behavior from those that maximize aggregate score.

The benchmark data, scorer, public baselines, and canonical results are publicly available.[1]

2 Related Work

Research on long-term conversational memory sits at the intersection of three lines of work: **long-context language modeling**, **memory-augmented assistant architectures**, and **benchmarks for persistent conversational memory**. ENGRAMA and ENGRAMABENCH are most directly related to systems that attempt to preserve user-relevant information across many interactions, but they are also informed by the broader question of whether memory should be represented as transcript inclusion, flat retrieval over extracted facts, or a more structured intermediate form.

2.1 Long-Context and Memory-Augmented Assistants

One increasingly strong baseline for long-term conversational tasks is simply to provide the model with as much prior history as possible at answer time. Improvements in long-context language models have made this strategy practical over progressively larger windows, and in many cases full-context prompting remains the most competitive fallback available. Its main strengths are simplicity and high fidelity to the original interaction record: the system does not need to decide in advance what to remember, because the transcript itself is treated as memory. At the same time, this strategy makes memory quality largely a function of context budget and prompt construction, and it scales poorly as histories become longer and

[1] https://github.com/julianacunadc/engramabench

more heterogeneous. It also offers relatively little control over how facts are organized once they leave the local discourse region in which they were originally introduced.

This limitation has motivated a growing class of **memory-augmented assistants** that explicitly index, compress, extract, or retrieve information from prior interactions. Prior systems such as **MemoryBank** [Zhong et al., 2023], **MemGPT / Letta** [Packer et al., 2023], **ReadAgent** [Lee et al., 2024], **PerLTQA** [Du et al., 2024], and MEM0 [Chhikara et al., 2025] represent different points in this design space. Some focus on extracting persistent user facts into a flat store; others partition long histories into retrievable memory units, maintain external memory buffers, or add agentic retrieval and reading stages around a language model. LongMemEval frames this design space in terms of indexing, retrieval, and reading stages [Wu et al., 2025], which is a useful abstraction for analyzing where long-term memory systems succeed or fail.

ENGRAMA can also be understood as a form of **retrieval-augmented generation (RAG)** in which the retrieval substrate is a structured memory graph rather than a flat document or chunk index. This structural perspective also relates to graph-based retrieval and synthesis systems such as **GraphRAG** [Edge et al., 2024] and **HippoRAG** [Gutiérrez et al., 2024], which explore how graph-structured retrieval substrates can support multi-hop recall and synthesis more effectively than flat retrieval alone. ENGRAMA is closest in spirit to this literature, but differs in one important respect. Rather than treating memory as either full transcript replay or flat semantic search over extracted snippets, it treats long-term conversational state as a **graph-structured memory surface**. The goal is not only to recover prior facts, but to preserve the relationships among spaces, entities, temporal traces, and cross-space associations so that later retrieval can operate over a more organized representation of user history. In that sense, ENGRAMA is motivated by the same broad problem as prior memory systems, while taking a more explicitly structural view of what conversational memory should look like.

2.2 Benchmarks for Long-Term Conversational Memory

Existing benchmarks have helped establish that long-term memory is a meaningful challenge for language-model-based assistants, but they vary substantially in what they measure. Early multi-session dialogue benchmarks such as **MSC** [Xu et al., 2022] and related long-horizon conversation datasets primarily emphasize coherence, persona consistency, or limited recall across a relatively small number of sessions. These benchmarks are useful for studying persistence in dialogue generation, but they do not isolate the kinds of delayed factual recall, abstention, temporal revision, and cross-space synthesis that arise in assistant settings.

More recent long-term benchmarks move closer to the setting we study here. **LoCoMo** introduces very long multi-session conversations grounded in personas and temporal event graphs, and evaluates models on memory-intensive tasks over those histories [Maharana et al., 2024]. Its core contribution is to show that even strong long-context or retrieval-augmented systems struggle when conversations become substantially longer and more entangled than the short-session settings common in prior work. **LongMemEval** takes a complementary step by focusing explicitly on chat assistants and defining a benchmark around multiple long-term memory abilities, including information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention [Wu et al., 2025]. It also contributes a useful decomposition of memory system design into indexing, retrieval, and reading, making it especially relevant as a methodological reference point.

Other benchmarks and task suites address adjacent aspects of long-term personalization, persistent memory, or question answering over chat history, including **MemoryBank** [Zhong et al., 2023], **PerLTQA** [Du et al., 2024], and related evaluations of personalized or long-horizon dialogue understanding. Collectively, these works establish two important facts: first, that memory failures remain common even for strong models; and second, that evaluating long-term memory requires more than simply increasing context length. However, they also leave several gaps.

In particular, prior benchmarks often underemphasize one or more of the following: **(i)** explicit integration across semantically distinct parts of a user's life, **(ii)** the interaction between temporal change and cross-session reasoning, **(iii)** abstention under fabricated or unsupported personal queries, and **(iv)**

synthesis questions whose answers are not stated verbatim in any single session. Some settings are also optimized either for human-human dialogue or for generic long-context understanding rather than for assistant-oriented memory systems that must accumulate and later operationalize structured user knowledge.

2.3 Positioning ENGRAMABENCH and ENGRAMA

ENGRAMABENCH is intended to complement rather than replace this prior work. Relative to LoCoMo, it places less emphasis on extremely long conversation length and multimodal grounded dialogue, and more emphasis on **assistant-style memory evaluation under explicit semantic-space structure**. Relative to LongMemEval, it is smaller in scale but sharper in one particular direction: it makes **cross-space reasoning** a first-class benchmark target and pairs it with adversarial abstention and emergent synthesis under controlled persona histories. Its design therefore trades breadth for interpretability. By organizing each persona around recurring semantic spaces and then authoring queries that selectively stay within, cross between, or reason over those spaces across time, the benchmark is able to distinguish memory behaviors that are often conflated under a single long-context score.

ENGRAMA itself is positioned analogously on the systems side. We do not present it as a universal replacement for full-context prompting, nor as a direct implementation of any single prior memory architecture. Instead, we use it as a structured counterpoint to two common alternatives: **brute-force transcript inclusion** and **flat retrieval over extracted memories**. This framing allows the paper to ask a narrower but more empirically tractable question: *when does structured memory provide a measurable advantage, and what tradeoffs does that advantage impose elsewhere?*

Taken together, the benchmark and the system are meant to contribute a more analyzable view of long-term conversational memory. Prior work has shown that long histories are difficult, that memory components can help, and that existing systems still underperform on sustained interaction. Our contribution is to isolate a regime in which **memory organization itself** becomes experimentally visible: cross-space reasoning over heterogeneous multi-session personal histories. That is the specific niche in which we position ENGRAMABENCH and ENGRAMA within the existing literature.

3 ENGRAMABENCH

We introduce ENGRAMABENCH, a benchmark for evaluating long-term conversational memory under multi-session interaction. The benchmark is designed to test not only whether a system can recover isolated facts from prior conversations, but also whether it can connect information across semantically distinct memory regions, reason over temporal change, refuse unsupported queries, and synthesize higher-level conclusions from dispersed evidence. In contrast to short-horizon personalization benchmarks, ENGRAMABENCH emphasizes *structured long-term recall*: information is distributed across many conversations, organized into distinct spaces of life or work, and often becomes useful only when combined across sessions and over time.

At a high level, each benchmark instance consists of a persona, a chronologically ordered set of conversations, and a set of evaluation queries written against that history. The benchmark is intended for systems that process conversations online, store some form of memory representation, and later answer questions about the accumulated interaction history. Although the benchmark can be used to evaluate any memory-augmented assistant, it is especially aimed at systems that must operate under realistic memory constraints rather than simply replaying the full transcript into the context window at answer time.

3.1 Problem Formulation

Each benchmark problem can be described as a tuple (H, q, a, E), where H is a chronologically ordered interaction history, q is a user query posed after that history, a is the reference answer or scoring target, and E is the annotated evidence supporting the query. The history H is composed of multiple conversations,

each associated with a timestamp, a conversation identifier, and a semantic space. Conversations are themselves multi-turn user-assistant exchanges.

The full `EngramaBench full_v1` benchmark contains **5 personas**, **100 conversations**, and **150 queries**. Each persona contributes **20 conversations** and **30 queries**. Queries are asked only after the underlying interaction history has been fully observed, so the task is not next-turn prediction but delayed long-term recall and reasoning over accumulated memory.

This setup captures a practical memory setting for personal assistants. A system must remember facts that may have been mentioned many sessions earlier, distinguish between information from different parts of a user's life, track updates to that information over time, and avoid answering unsupported questions. The benchmark therefore mixes direct factual recall with cross-space integration, temporal reasoning, and abstention.

3.2 Dataset Construction

ENGRAMABENCH is organized around five canonical personas: **Priya Sharma**, **Carlos Mendoza**, **Sam Torres**, **Diane Chen**, and **Kai Nakamura**. Each persona is defined over five recurring semantic spaces that partition the major domains of that person's ongoing life. For example, Priya's spaces include `lab`, `publications`, `grants`, `teaching`, and `collaboration`, while Carlos's include `investors`, `product`, `hiring`, `market`, and `personal`. This space structure is important: it creates a memory landscape in which some questions are resolvable within a single region, while others require combining evidence across multiple regions.

The five personas were chosen to cover different professional domains, life stages, and discourse styles, reducing the risk that benchmark performance is driven by overfitting to a single user archetype. The resulting set spans academic research, startup operations, creative practice, family-career coordination, and graduate study, which in turn induces different kinds of memory demands over numbers, plans, relationships, temporal change, and emotional context.

For each persona, the benchmark includes 20 timestamped conversations arranged in chronological order. The conversations collectively trace a medium-horizon narrative spanning late 2025 into early 2026. These conversations are synthetically authored through an LLM-assisted generation pipeline constrained by persona blueprints, cross-space threads, and temporal arcs, and are then checked for structural and semantic consistency.

Persona blueprints specify not only spaces but also higher-level cross-space threads and temporal arcs, such as grant pressure spilling into lab management, fundraising affecting burnout, or academic stress interacting with social withdrawal. These arcs are later used during query authoring to ensure that the benchmark contains meaningful long-range dependencies rather than disconnected trivia.

Query generation follows a fixed per-persona distribution implemented in the benchmark pipeline: **6** `single_space` queries, **6** `cross_space` queries, **6** `temporal_cross_space` queries, **4** `emergent_insight` queries, and **8** `adversarial` queries, for a total of **30** per persona. Most queries are generated with model assistance from the canonical conversation set and then constrained by explicit evidence requirements. Adversarial queries are generated separately from persona-specific false-premise traps and are designed to test refusal behavior.

3.3 Query Taxonomy

ENGRAMABENCH evaluates five task families:

- `single_space`: factual recall within a single semantic space. These questions test whether the system can recover a specific entity, number, date, or short span grounded in one region of memory; e.g., *"What is the deadline for Priya's R01 proposal?"*
- `cross_space`: integration across two or more spaces. These questions require linking information that is distributed across distinct domains of the persona's life; e.g., *"What paper was crucial for the R01 preliminary data section despite being rejected by Nature Methods?"*

- `temporal_cross_space`: temporal reasoning over information distributed across spaces and time. These questions ask how a state, metric, or situation changed and require the system to reason over chronology rather than merely retrieve isolated facts; e.g., *"How did Carlos's MRR growth trajectory change from his first Kaszek meeting to where it is now?"*
- `adversarial`: fabricated or false-premise questions for which the correct behavior is abstention; e.g., *"What phase is Priya's clinical trial currently in?"*
- `emergent_insight`: synthesis questions whose best answers are not stated verbatim in any single conversation, but instead require combining multiple pieces of evidence into a higher-level conclusion; e.g., *"What were the key interconnected factors that enabled the successful R01 submission and strong scoring?"*

This taxonomy is meant to separate memory abilities that are often conflated. A system may be strong at local factual recovery yet weak at cross-space integration; it may answer many direct questions correctly but fail on temporal comparison; or it may retrieve broadly but perform poorly on unsupported questions. By reporting each family separately, the benchmark exposes these tradeoffs directly.

3.4 Evidence and Scoring

Each query is accompanied by explicit evidence annotations. In particular, query records include `evidence_conversation_ids` and `evidence_message_ids`, which identify the supporting conversations and messages from the canonical history. During query generation, the benchmark pipeline validates that evidence exists, that cited messages belong to the cited conversations, and that the evidence spans all declared required spaces for multi-space queries. This makes the benchmark auditable at the artifact level even when final scoring is based on answer quality rather than retrieved evidence alone.

The scorer supports seven answer types: `entity`, `date`, `number`, `set`, `short_span`, `abstain`, and `insight`. Entity answers are normalized through an alias registry; dates allow limited fuzzy matching; numbers use tolerance-based matching; sets are scored with set F1; short spans use token-level F1; abstention is binary; and insight questions currently use token-level F1 over the reference answer as a lightweight proxy rather than a full judge-based rubric. These answer-type scorers are aggregated into the five task-family metrics reported throughout the paper.

At the top level, ENGRAMABENCH reports the following family scores: `single_space_f1`, `cross_space_f1`, `temporal_cross_space_f1`, `adversarial_accuracy`, and `emergent_insight_score`. The overall `composite_score` is a weighted mixture of these components. Let

$$F = \frac{1}{3}(S_{\text{single}} + S_{\text{cross}} + S_{\text{temporal}})$$

be the mean of the three factual families. The composite score is then computed as

$$S_{\text{composite}} = 0.5F + 0.25S_{\text{adversarial}} + 0.25S_{\text{emergent}}.$$

This weighting gives equal total weight to factual memory, adversarial abstention, and higher-level synthesis while preventing any single factual slice from dominating the benchmark.

We therefore interpret `emergent_insight` scores somewhat more cautiously than the factual slices. The current scorer implementation captures overlap with a reference synthesis answer, but it does not yet evaluate correctness, specificity, and cross-evidence grounding with a dedicated judge model.

3.5 Benchmark Limitations

ENGRAMABENCH is reproducible and stable at the run-artifact level: the benchmark is tied to fixed conversation snapshots, frozen query files, explicit evidence annotations, versioned registries, and deterministic scoring outputs. This makes it well suited for comparing systems, baselines, and ablations under a common evaluation surface.

At the same time, the benchmark does not yet fully audit runtime evidence provenance at the exact `message_id` or `chunk_id` level for every evaluated system. In other words, the current benchmark tells us whether the final answer is correct and whether a gold evidence path exists in the source data, but not always whether the system actually retrieved the exact gold evidence during inference. We therefore treat ENGRAMABENCH as a reliable benchmark for *system-level comparison and ablation analysis*, while reserving stronger claims about fine-grained retrieval provenance for future instrumentation.

4 ENGRAMA

ENGRAMA is a proprietary **graph-structured long-term memory system** for conversational retrieval and reasoning. The system is designed for settings in which an assistant must process user-assistant interactions incrementally, retain information over many sessions, and later answer questions that may depend on facts distributed across distinct semantic regions of a user's history. Rather than treating memory as a flat store of extracted snippets or summaries, ENGRAMA organizes accumulated interaction traces into a structured representation intended to preserve cross-space associations, temporal continuity, and entity-level coherence.

This section describes ENGRAMA at a high level. We focus on the aspects needed to interpret the empirical results in the benchmark, while omitting implementation-specific retrieval heuristics, production orchestration details, and internal scoring formulas for confidentiality reasons.

4.1 System Overview

At a high level, ENGRAMA processes a stream of timestamped user-assistant conversations and maintains a graph-structured memory over that stream. The memory is organized around four broad classes of information: **entities**, **semantic spaces**, **temporal traces**, and **associative links**. Entities provide identity continuity across mentions; spaces separate different regions of the user's life or work; temporal traces preserve the order and recency of events; and associative links connect related pieces of information that may become jointly useful at answer time. In particular, associative links capture relationships that span space boundaries, such as connecting a person mentioned in a grants conversation to their role in a collaboration space, or tying a workload signal in one area of life to a consequence that emerges in another.

This design is motivated by a core observation reflected in ENGRAMABENCH: many difficult questions are not solved by locating a single memorized fact. Instead, they require recovering a *configuration* of related memories. For example, a question may ask about a grant decision whose answer depends jointly on activity in a grants space, a publication space, and a collaboration space, or about a current situation that only becomes interpretable when compared against earlier sessions. A graph representation offers a natural substrate for this kind of retrieval because it supports activation over relations and neighborhoods rather than only over isolated text chunks.

At inference time, ENGRAMA maps a query into a region of the memory graph, retrieves a focused subset of relevant memory, and constructs an answer from that subset. The system therefore sits between two common alternatives. Relative to full-context prompting, it avoids replaying the entire transcript at answer time. Relative to flat vector retrieval over extracted memories, it retains a richer notion of structure among remembered items. Figure 1 illustrates this pipeline at a high level.

4.2 Ingestion and Memory Representation

ENGRAMA ingests interaction history incrementally. Each new conversation is processed into a structured memory update rather than stored only as raw text. At a conceptual level, this update identifies salient entities, attributes, relations, and contextual signals from the conversation and attaches them to the existing memory graph. The resulting representation preserves both the local content of the new exchange and its relationship to previously observed parts of the user's history.

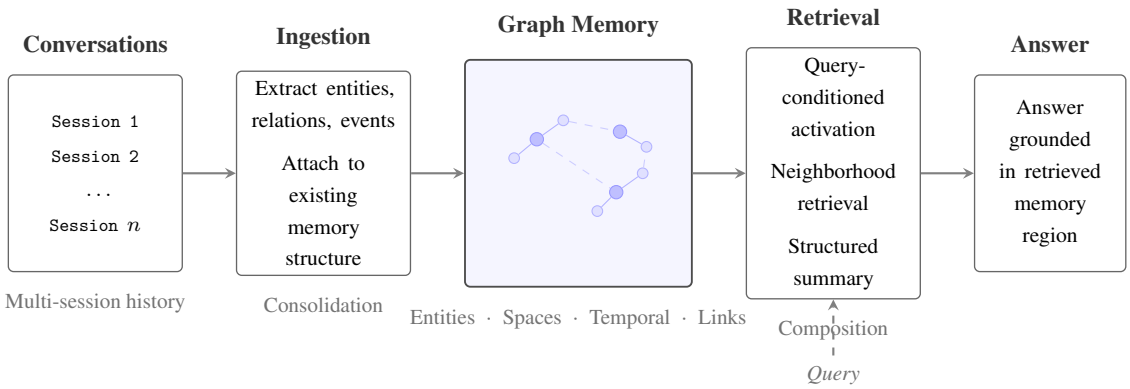

Figure 1: High-level overview of ENGRAMA. Conversations are incrementally consolidated into a graph-structured memory organized around entities, semantic spaces, temporal traces, and cross-space associative links (dashed edges). At query time, the system activates a relevant neighborhood rather than replaying the full transcript, converts it into a structured summary, and generates an answer grounded in that region. This graph-based retrieval path is what enables ENGRAMA's cross-space advantage over flat retrieval and full-context prompting.

Two properties of the representation are particularly important for the benchmark setting. First, ENGRAMA retains **space information** as a first-class part of memory. This allows the system to distinguish, for example, a user's work commitments from family obligations or creative practice from financial logistics, while still permitting interactions across those regions. Second, ENGRAMA retains **temporal information** associated with remembered content. This is essential for update-sensitive and temporally comparative questions, where the answer depends not only on what was said but also on when it was said relative to other events.

The graph structure also allows repeated references to reinforce previously seen entities or situations rather than creating a new disconnected memory each time they are mentioned. In effect, ENGRAMA attempts to accumulate a structured memory state over time, rather than a sequence of independent retrieved passages. This helps explain why the system performs relatively strongly on `cross_space` questions in ENGRAMABENCH: those questions often depend on connections that are preserved naturally in a graph-structured representation.

4.3 Retrieval and Composition

When answering a query, ENGRAMA performs retrieval in two broad stages: **memory activation** and **answer-time composition**. The first stage identifies a relevant subregion of the graph, while the second converts the retrieved subgraph into a response-compatible context. Conceptually, the retrieval process is guided by the entities, spaces, and temporal cues implied by the query, and then expands outward to collect additional memory that is likely to be jointly useful.

This retrieval procedure is designed to balance precision and compositionality. If retrieval is too narrow, the system may recover only a local fact and miss the broader configuration needed for a cross-space or temporal answer. If it is too broad, the answering model receives diffuse context and becomes more likely to paraphrase, hedge, or anchor on irrelevant evidence. ENGRAMA addresses this by retrieving *structured memory regions* rather than only nearest-neighbor memory fragments. In practice, the downstream answering stage receives a focused, structured summary of the retrieved memory rather than the full graph itself.

Answer construction is therefore not a separate external module layered on top of raw retrieval, but the final stage of a memory pipeline. The system first identifies what region of memory appears relevant, then organizes that region into an answer-oriented representation, and finally emits a natural-language answer. This framing is especially important for understanding the benchmark results: several observed

failures are not pure retrieval misses, but rather failures to convert a partially correct memory state into the exact answer form expected by the task.

4.4 Configurable Design Levers

The experiments in this paper study two configurable design levers in ENGRAMA.[2]

Lever 1 (L1) combines **entity-first activation** with a **query planner**. At a conceptual level, this lever biases retrieval toward memory regions that appear structurally aligned with the entities and semantic intent of the query. Rather than treating the query as an undifferentiated text string, L1 attempts to expose a more memory-aware retrieval path by foregrounding likely anchors and relevant spaces before graph activation fully unfolds.

Lever 3 (L3) is a **typed answer layer**. Its role is not to change what the system remembers, but to change how strongly retrieved evidence is converted into an explicit answer candidate when the task appears to demand a specific answer form. Conceptually, L3 is meant to reduce failures in which the system retrieves relevant evidence but emits a vague or weakly structured response instead of a directly scorable answer.

These levers are useful analytically because they separate two common sources of failure in long-term memory systems: (1) retrieving the wrong memory region, and (2) failing to convert a roughly correct memory state into a precise output. The ablation results in Section 7 show that both levers affect ENGRAMA's behavior in nontrivial ways. In particular, they help preserve cross-space performance, but in their current implementations they do not improve global composite score. This supports the view that memory system quality depends not only on what is stored, but also on how queries are routed through memory and how retrieved evidence is rendered into an answer.

5 Experimental Setup

This section describes the benchmark protocol, the evaluated systems, the ablation settings, and the reporting conventions used in the experiments.

5.1 Benchmark Protocol

All reported results use the `EngramaBench full_v1` benchmark described in Section 3. The benchmark contains **5 personas**, **100 conversations**, and **150 queries** in total. Each persona contributes 20 conversations and 30 queries, with the fixed per-persona query distribution of 6 `single_space`, 6 `cross_space`, 6 `temporal_cross_space`, 4 `emergent_insight`, and 8 `adversarial` queries. Unless otherwise stated, scores are computed with the **v1.4.0** scorer using the **unified entity registry** setting.

For ENGRAMA, each query is evaluated against a clean clone of a persona-specific canonical memory snapshot. This avoids cross-query contamination and ensures that benchmark runs measure question answering over a fixed memory state rather than a cumulatively mutated one. For non-native-memory baselines such as GPT-4o full-context and MEM0, the same persona histories and query files are used, but without graph snapshotting because those systems do not maintain ENGRAMA's internal state representation.

5.2 Systems and Baselines

All three evaluated systems use `gpt-4o-2024-08-06` at **temperature 0.3** as the answering model. This ensures that benchmark differences primarily reflect memory and retrieval architecture rather than differences in the underlying language model.

We evaluate three system families.

[2] An additional lever, **L2**, exists in the broader system architecture as an intermediate evidence-structuring layer, but it was disabled in all benchmark runs reported in this paper and is therefore excluded from the ablation analysis.

GPT-4o full-context is the primary brute-force baseline. For each query, all persona conversations are concatenated in chronological order and injected directly into the prompt, with no memory compression and no retrieval stage.

MEM0 is an open-source memory baseline using vector retrieval over extracted memories. We seed MEM0 from the canonical conversation histories, store extracted memories in ChromaDB, retrieve the top-k memories at question time, and then answer using only the retrieved memories as context. The reported MEM0 runs use `text-embedding-3-small` for embeddings and top-$k = 20$ retrieval at query time.

ENGRAMA is evaluated in its benchmark configuration as a proprietary graph-structured memory system. Queries are run against persona-specific canonical snapshots, with retrieval and answer construction mediated by ENGRAMA's internal graph memory pipeline rather than full-context prompting or flat vector search.

5.3 Ablations

In addition to the full ENGRAMA configuration, we report three ablations:

- `no-L1`: disables Lever 1, which combines entity-first activation and query planning.
- `no-L3`: disables Lever 3, the typed answer layer.
- `no-L1-L3`: disables both Lever 1 and Lever 3 simultaneously.

These ablations are designed to isolate two broad classes of effect: routing the query toward a more structured memory region, and converting retrieved evidence into a more directly scorable answer form.

5.4 Metrics and Reporting

We report the five task-family metrics defined in Section 3: `single_space_f1`, `cross_space_f1`, `temporal_cross_space_f1`, `adversarial_accuracy`, and `emergent_insight_score`, together with the aggregate `composite_score`. The composite score is computed directly from the benchmark scorer using the weighting described in Section 3.

In addition to global results, we also analyze per-persona composite scores in order to expose variation across user archetypes. This is important because some systems are relatively stable across personas while others are much more sensitive to persona domain, narrative style, or temporal complexity.

5.5 Cost Accounting

We report **query-time cost** using frozen pricing tables in the evaluation runners. For GPT-4o full-context and MEM0, this corresponds to the token cost of answer generation at query time. For ENGRAMA, this corresponds to the query-time answering cost under the benchmark runner. These figures are intended to compare online serving cost under fixed benchmark conditions rather than full end-to-end system cost.

This distinction matters especially for MEM0. The reported MEM0 query cost excludes the offline cost of seeding the memory store, including memory extraction and embedding generation, and therefore should be interpreted as **online query cost only**. Likewise, the ENGRAMA figures do not attempt to amortize the cost of building canonical snapshots. We treat cost primarily as a practical serving-side comparison between three answer-time regimes: brute-force full-context prompting, graph-structured memory retrieval, and vector retrieval over extracted memories.

6 Main Results

This section reports the main comparative results on `EngramaBench full_v1`. We focus on the primary system comparison among **GPT-4o full-context**, **ENGRAMA full**, and **MEM0**. Ablated ENGRAMA variants are analyzed separately in Section 7.

System	single	cross	temporal	advers.	emergent	Composite	Cost
GPT-4o full-context	**0.7339**	0.6291	**0.3902**	0.9750	**0.3305**	**0.6186**	$3.33
ENGRAMA full	0.5997	**0.6532**	0.3230	0.8000	0.2963	0.5367	$0.67
MEM0	0.2848	0.5266	0.2356	**1.0000**	0.2255	0.4809	$0.36

Table 1: Main results on ENGRAMABENCH `full_v1` (150 queries, 5 personas). All three systems use GPT-4o as the answering model; differences therefore reflect memory architecture, not model quality. GPT-4o full-context leads globally, but ENGRAMA achieves the highest `cross` score—the only metric where structured memory outperforms brute-force context inclusion. MEM0 is cheapest but weakest across all factual families. Cost is total query-time cost across 150 queries. Ablated ENGRAMA variants are reported separately in Table 3. Bold indicates best per column.

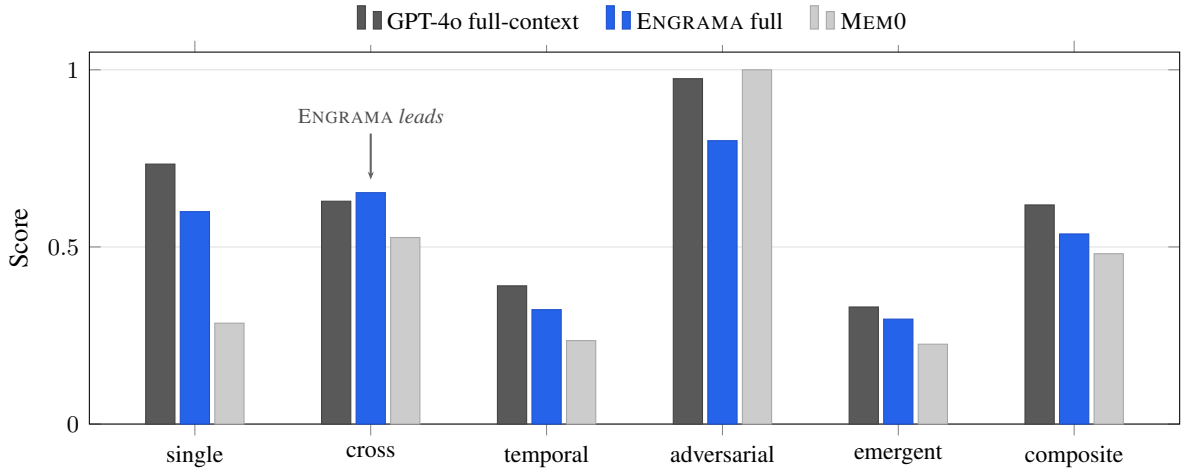

Figure 2: Per-metric comparison of the three primary systems on ENGRAMABENCH (150 queries). GPT-4o full-context (dark) leads on most metrics and overall composite. ENGRAMA (blue) is the only system to outperform full-context prompting, doing so on `cross`—the metric most diagnostic of structured long-term memory. MEM0 (light) is weakest on factual families but achieves perfect adversarial abstention. Exact values in Table 1.

6.1 Overall Comparison

Table 1 shows that **GPT-4o full-context** achieves the highest global score on the benchmark, reaching a **composite score of 0.6186**. **ENGRAMA full** scores **0.5367**, while **MEM0** reaches **0.4809**. Because all three systems use the same answering model, these differences reflect the quality of the memory and retrieval pipeline rather than differences in the underlying language model.

At the metric level, GPT-4o full-context is strongest on `single_space` (**0.7339**), `temporal_cross_space` (**0.3902**), and `emergent_insight` (**0.3305**), and it also performs very strongly on `adversarial` (**0.9750**). MEM0, by contrast, is weakest overall, especially on `single_space` (**0.2848**) and `temporal_cross_space` (**0.2356**), though it achieves perfect adversarial abstention (**1.0000**). ENGRAMA full sits between the two baselines globally, but with a qualitatively different profile: it leads on the benchmark slice most directly aligned with structured memory behavior.

The performance gap between MEM0 and the other two systems is notable: even with the same answering model, flat vector retrieval over extracted memories is not sufficient to close the distance. This suggests that the bottleneck is not retrieval per se, but the quality and structure of what is stored.

6.2 Cross-Space Advantage

The clearest signature of ENGRAMA's design appears in `cross_space` performance. **ENGRAMA full** achieves **0.6532**, compared to **0.6291** for GPT-4o full-context and **0.5266** for MEM0. This makes

Persona	ENGRAMA	GPT-4o	Δ
Priya (PI / professor)	0.5716	0.6066	−0.0350
Carlos (founder)	0.5294	0.6397	−0.1103
Sam (artist / musician)	0.5569	0.6405	−0.0836
Diane (product lead)	0.5329	0.6052	−0.0723
Kai (PhD student)	0.4928	0.6037	−0.1109
Mean	**0.5367**	**0.6191**	**−0.0824**

Table 2: Per-persona composite scores for ENGRAMA full vs. GPT-4o full-context. Kai is consistently the hardest persona for ENGRAMA; the gap is smallest for Priya.

ENGRAMA the strongest system on the only task family that explicitly requires connecting information across semantically distinct regions of a user's history—and as Table 3 shows, it remains the highest-scoring configuration even when compared to its own ablated variants.

This result matters because `cross_space` is the benchmark family most diagnostic of structured long-term memory. A strong `single_space` score can come from local recall or brute-force transcript access; a strong `cross_space` score requires something more compositional. The system must identify which parts of memory belong together even when the relevant evidence is not co-located in a single conversation or space. ENGRAMA's graph-structured memory appears to help precisely in this setting, preserving associations that flat retrieval methods or brute-force full-context prompting do not exploit as effectively.

The contrast with MEM0 is especially informative. MEM0 retrieves top-k extracted memories from a vector store and answers with the same LLM used elsewhere, yet it trails ENGRAMA by **12.7 points** on `cross_space` (**0.5266** vs. **0.6532**). This gap suggests that the difference is not just retrieval versus no retrieval, but **structured retrieval versus flat retrieval**. In other words, what matters is not only whether a system retrieves, but how memory is structured for retrieval.

6.3 Cost-Quality Tradeoff

The three systems also occupy clearly different positions in the cost-quality landscape. Across all 150 benchmark queries, **GPT-4o full-context** costs approximately **$3.33**, **ENGRAMA full** approximately **$0.67**, and **MEM0** approximately **$0.36** in query-time cost. GPT-4o therefore offers the best quality but at the highest serving cost; MEM0 is the cheapest query-time baseline but also the weakest in quality; and ENGRAMA occupies the middle ground.

ENGRAMA reaches about **86.7%** of GPT-4o's composite score at roughly **20%** of the query-time cost, while still outperforming GPT-4o on `cross_space`. MEM0 is cheaper still, but the quality drop is much larger: it reaches only **77.7%** of GPT-4o's composite and falls meaningfully behind ENGRAMA on factual recall, temporal reasoning, and cross-space integration. Taken together, these results suggest that ENGRAMA offers the most compelling tradeoff among the evaluated memory-augmented systems: it does not match the absolute accuracy of full-context prompting, but it recovers much of that performance at substantially lower cost and with a targeted advantage in cross-space reasoning.

One caveat is important here. The reported MEM0 cost is **query-time only** and excludes offline memory seeding, extraction, and embedding generation. We therefore interpret the cost numbers as serving-time comparisons rather than full end-to-end system costs. Even under that narrower lens, ENGRAMA's additional query-time cost over MEM0 is accompanied by a substantial gain in answer quality.

6.4 Per-Persona Variation

Per-persona results (Table 2) reveal another important pattern: the systems differ not only in average performance, but also in **stability across user archetypes**. GPT-4o full-context is the most stable system, with per-persona composite scores ranging from **0.6037** (Kai) to **0.6405** (Sam), a spread of only **0.0368**.

Variant	Δcomposite	Δsingle	Δcross
no-L1	+0.0121	+0.0333	−0.0164
no-L3	+0.0134	+0.0325	−0.0499
no-L1-L3	+0.0000	+0.0325	−0.0546

Table 3: Ablation deltas relative to ENGRAMA full (composite = 0.5367). The pattern reveals a design tension: removing L1 or L3 individually improves global composite but degrades `cross_space`, the metric most diagnostic of structured memory. Removing both eliminates the composite gain while further degrading cross-space, indicating that the levers interact sub-additively on composite but complementarily on cross-space.

ENGRAMA full is less stable but still reasonably consistent, ranging from **0.4928** (Kai) to **0.5716** (Priya), for a spread of **0.0788**. MEM0 is the most variable system by far, ranging from **0.3687** (Kai) to **0.5563** (Diane), for a spread of **0.1876**.

This instability is useful analytically. It suggests that flat memory extraction and retrieval are much more sensitive to persona domain, discourse style, and temporal structure than either full-context prompting or graph-structured memory. In particular, **Kai** is consistently the hardest persona for the memory baselines: ENGRAMA posts its lowest persona score on Kai, and MEM0 drops even more sharply there. This persona contains substantial interaction between research, coursework, wellness, social life, and career preparation, making it especially demanding for systems that must connect signals across domains and over time. By contrast, Diane is comparatively strong for both ENGRAMA and MEM0, suggesting that some benchmark narratives are easier to compress into stable memory representations than others.

The per-persona analysis reinforces a practical takeaway: full-context prompting is the most robust approach, likely because it avoids the compression and routing errors that memory systems can accumulate. ENGRAMA narrows this robustness gap substantially compared to MEM0, but the variation across personas suggests that memory stability across diverse user histories remains an open challenge.

7 Ablations and Analysis

The main comparison establishes *what* happens: ENGRAMA trails GPT-4o globally but leads on `cross_space`. The ablation study addresses *why*. It reveals that the components driving the cross-space advantage are not the same ones that maximize composite score, creating a nontrivial design tension between specialization and aggregate optimization.

We evaluate four ENGRAMA variants: the full system, `no-L1`, `no-L3`, and `no-L1-L3`. Here, **L1** combines entity-first activation with query planning, while **L3** is the typed answer layer described in Section 4.

7.1 Ablation Matrix

Table 3 summarizes the ablation results relative to the full ENGRAMA configuration. The full system achieves **0.5367** composite, with the strongest `cross_space` score among ENGRAMA variants at **0.6532**. Removing **L1** raises composite to **0.5488**. Removing **L3** raises it further to **0.5501**, the highest composite among all ENGRAMA variants. However, both of these gains come with declines in `cross_space`: `no-L1` falls to **0.6368**, and `no-L3` falls more sharply to **0.6033**. Removing both levers simultaneously yields **0.5367**, exactly matching the full system, while further reducing `cross_space` to **0.5986**.

The matrix therefore does not support a simple interpretation in which L1 and L3 are either purely helpful or purely harmful. Instead, the results indicate a patterned tradeoff. Relative to full ENGRAMA, the ablations increase `single_space` and, in the case of `no-L1`, also modestly increase `temporal_cross_space`; yet these gains are accompanied by a loss of the very behavior most characteristic of ENGRAMA as a structured memory system, namely cross-space integration.

7.2 Lever Tradeoffs

Viewed individually, both levers appear to exchange some amount of global composite for better cross-space behavior.

Removing **L1** improves overall composite by **+0.0121**, raises `single_space` from **0.5997** to **0.6330**, and raises `temporal_cross_space` from **0.3230** to **0.3397**. At the same time, it lowers `cross_space` from **0.6532** to **0.6368**. This suggests that L1 is doing something structurally useful for routing queries toward the right memory region, but that the same routing bias may also introduce overhead or mistakes that hurt aggregate performance outside the cross-space regime.

Removing **L3** produces a similar but even sharper pattern. Composite increases by **+0.0134**, and `single_space` increases from **0.5997** to **0.6322**. Yet `cross_space` drops by nearly **5 points**, from **0.6532** to **0.6033**. This indicates that L3 is not merely an output formatting convenience. Rather, it appears to help ENGRAMA convert partially structured memory states into answers that preserve cross-space evidence chains more effectively, even though that same mechanism does not improve the benchmark as a whole in its current implementation.

Taken together, these results suggest that L1 and L3 are not generic quality multipliers. They are more specialized components whose value is concentrated in the portion of the benchmark that most strongly rewards structured retrieval and compositional memory use.

7.3 Interaction Effect

The most revealing result in the ablation matrix is `no-L1-L3`. If L1 and L3 simply imposed independent overhead, then removing both should have produced a larger gain than removing either one alone. That is not what happens. Instead, `no-L1-L3` returns to **0.5367**, exactly matching the full system, while preserving the `single_space` gains and losing even more of the `cross_space` advantage.

This behavior indicates a real **interaction effect** between the levers. A more precise interpretation is that **L1 and L3 are each beneficial in isolation relative to the `no-L1-L3` baseline on global composite, but their gains do not compose when both are enabled simultaneously**. In contrast, on `cross_space`, the two levers appear complementary: the full system outperforms either isolated lever and substantially exceeds the `no-L1-L3` baseline. Taken together, these results suggest that the interaction between L1 and L3 is objective-dependent: the pair is better aligned with cross-space specialization than with the benchmark's global composite objective.

This interaction also helps explain the difference between earlier pilot results and the full benchmark. In the smaller pilot benchmark, the benefits of these mechanisms appeared larger and more uniformly positive. In the full benchmark, which contains more heterogeneous query families and more opportunities for local retrieval or answer-construction failure, their costs become more visible. The full benchmark therefore exposes a more realistic system-level tradeoff: the mechanisms that make ENGRAMA distinctive are also the ones that currently prevent it from maximizing global composite.

7.4 Failure Modes

The ablation study, together with the main benchmark results, highlights several recurring failure modes.

First, **temporal reasoning remains difficult for all systems**, including ENGRAMA. Even the strongest model on this slice, GPT-4o full-context, reaches only **0.3902**, while ENGRAMA full reaches **0.3230**. This suggests that time-sensitive reasoning over multi-session conversational memory is still a broadly unresolved problem rather than a weakness of any one architecture.

Second, ENGRAMA exhibits a recurring tension between **structured retrieval** and **typed answer precision**. Some failures arise because the system retrieves the right memory region but emits a poorly formatted answer; others arise because the query activates a structurally plausible but suboptimal subgraph. The divergent effects of L1 and L3 on cross-space versus composite support this interpretation.

Third, the broader comparison highlights that preserving exact facts through memory construction is itself a core challenge. The benchmark tests the full pipeline from ingestion to retrieval to answer

realization, not retrieval alone.

Fourth, ENGRAMA's lower adversarial abstention (**0.8000**, vs. **0.9750** for GPT-4o and **1.0000** for MEM0) reflects a structural property of graph-based retrieval: multi-hop activation can surface evidence that is structurally plausible but factually irrelevant to fabricated queries, leading the answering model to respond rather than abstain. Systems with less retrievable context are naturally more conservative.

Taken together, these failure modes suggest that ENGRAMA's design is not overbuilt—it is **misaligned with the global objective in a structured way**. The benchmark is surfacing meaningful architectural tradeoffs rather than merely rewarding larger context windows.

8 Discussion and Limitations

What do the results tell us about the role of memory structure? The benchmark exposes a regime in which memory organization itself changes what the system can do. If the benchmark rewarded only local factual recall, structured memory would offer limited conceptual benefit. Cross-space queries instead require recovering configurations of related facts that were never co-located in a single conversation, and here the graph representation provides a measurable edge.

The coexistence of this advantage with a lower global composite is itself informative. It means that structured memory is not yet a drop-in replacement for full-context prompting, but it is already a sharper tool for specific reasoning demands that transcript inclusion handles only incidentally.

The ablation results sharpen this point. Current memory architectures face a coordination problem: the mechanisms that help organize memory for cross-space use are not automatically the ones that maximize aggregate utility. This suggests that the next frontier is not simply more memory structure, but **more selective** memory structure—stronger when it is needed, quieter when it is not. In that sense, the benchmark is doing what it should: surfacing the engineering tradeoffs that determine how memory systems actually behave, rather than merely ranking them by a single score.

The comparison with MEM0 sharpens this interpretation. MEM0 is cheaper at query time and extremely conservative on unsupported questions, but it loses substantial ground on factual precision, temporal reasoning, and cross-space composition. This suggests that a memory system's bottleneck is not only retrieval over stored items, but the entire pipeline by which conversational experience is compressed into reusable memory. Put differently, *what is remembered* and *how it is structured* are inseparable design questions. ENGRAMA's advantage over MEM0 on `cross_space`, despite using the same answering model, is consistent with this view.

Several limitations remain.

First, **ENGRAMABENCH is built from synthetic but tightly constrained conversations**, not naturally occurring human-assistant logs. We regard this as an acceptable design choice for controlled benchmarking—canonical personas, evidence annotations, and known cross-space arcs would be difficult to obtain from organic logs—but the benchmark should not be interpreted as a direct measurement of production behavior.

Second, the benchmark is reproducible at the run-artifact level, yet it does **not fully audit runtime evidence provenance** at the exact `message_id` or `chunk_id` level for every evaluated system. Current results support strong claims about comparative answer quality, but more limited claims about exact evidence recovery dynamics.

Third, some ability regions remain underexplored. Temporal reasoning is difficult for all systems, including full-context prompting, suggesting that the current query formulation captures a genuinely hard problem. The `emergent_insight` score is implemented as token-level F1 over a reference answer rather than a judge-based rubric, and should be interpreted more cautiously than the factual slices.

Finally, there is a deliberate **disclosure boundary** in this paper. ENGRAMA is a proprietary system, and we therefore describe its architecture only at the level needed to make the empirical claims interpretable. We report its structural design, benchmark behavior, and ablation surfaces, but omit implementation-specific retrieval heuristics, graph weighting formulas, internal prompts, and production orchestration

details. We view this as a reasonable scientific compromise: sufficient transparency to support evaluation and analysis, without treating the paper as an implementation manual.

Taken together, these limitations suggest a productive future agenda rather than a weakening of the present result. The benchmark should be extended with finer-grained provenance instrumentation and possibly additional real-world or semi-naturalistic dialogue settings. The system should be improved so that its structural strengths translate more consistently into global quality. And the broader research question remains open: not whether long-term memory matters, but **what kinds of memory representation produce the best tradeoff between quality, cost, and robustness as conversational histories become more heterogeneous over time**.

9 Conclusion

We introduced ENGRAMABENCH, a benchmark for long-term conversational memory, and used it to evaluate ENGRAMA against GPT-4o full-context and MEM0. The main empirical finding is that memory representation changes system behavior in measurable ways: ENGRAMA's graph structure enables a cross-space advantage that neither full-context prompting nor flat vector retrieval achieves, even though full-context prompting remains the strongest general-purpose strategy.

The ablations reveal that this advantage comes with a real engineering cost: the components that drive cross-space strength do not simultaneously maximize aggregate score. This is not a failure of the system but a **design frontier**—one that structured memory architectures will need to navigate as they mature.

More broadly, our results argue for a shift in how long-term conversational memory is evaluated. The key question is not only whether a system can remember more of the past, but whether it can remember it in a form that remains useful across heterogeneous user narratives, changing temporal contexts, and multi-space reasoning demands. ENGRAMABENCH is intended as a step toward measuring that ability directly.

Future work should proceed in three directions. First, the benchmark should be extended with finer-grained evidence provenance and stronger evaluation for synthesis-heavy answers. Second, memory architectures like ENGRAMA should be improved so that their structural strengths translate more consistently into global benchmark gains. Third, additional baselines and more naturalistic conversational settings are needed to better characterize the boundary between brute-force context inclusion and genuinely effective long-term memory systems.

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